

5.3.16 Extended interrupt and event controller input (EXTI) characteristics

The pulse on the interrupt input must have a minimal length in order to guarantee that it is detected by the event controller.

Table 51. EXTI input characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
PLEC	Pulse length to event controller	-	20	-	-	ns

5.3.17 12-bit analog-to-digital converter ADC characteristics

Unless otherwise specified, the parameters given in Table 52 are values derived from tests performed under ambient temperature, f_{HCLK} frequency, and V_{DDA} supply voltage conditions summarized in Table 16.

Note: It is recommended to perform a calibration after each power-up.

Table 52. 12-bit ADC characteristics

Specified by design - Not tested in production.

Symbol	Parameter	Conditions						Min	Typ	Max	Unit
V _{DDA}	Analog power supply for ADC ON	-						2.70	-	3.6	V
V _{REF+}	Positive reference voltage	-						2.5	-	V _{DDA}	
V _{REF-}	Negative reference voltage	-						V _{SSA}			
f _{ADC}	ADC clock frequency	2.7V ≤ V _{DDA} ≤ 3.6 V						8	-	36	MHz
f _S with R _{AIN} =47 Ω and C _{PCB} = 22 pF	Sampling rate for slow channels	Resolution = 12 bits	All modes	2.7 V ≤ V _{DDA} ≤ 3.6 V	-40°C ≤ T _J ≤ 140°C	f _{ADC} = 36 MHz	SMP = 2.5	-	2.25	-	MSPS
		Resolution = 10 bits						-	2.57	-	
		Resolution = 8 bits						-	3	-	
		Resolution = 6 bits						-	4.5	-	
t _{TRIG}	External trigger period	Resolution = 12 bits						16	-	-	1/f _{ADC}
V _{AIN}	Conversion voltage range	-						0	-	V _{REF+}	V
R _{AIN} ⁽¹⁾	External input impedance	Resolution = 12 bits, T _J = 140°C						-	-	110	Ω
		Resolution = 12 bits, T _J = 125°C						-	-	610	
		Resolution = 10 bits, T _J = 140°C						-	-	2305	
		Resolution = 10 bits, T _J = 125°C						-	-	4290	
		Resolution = 8 bits, T _J = 140°C						-	-	11110	
		Resolution = 8 bits, T _J = 125°C						-	-	15860	
		Resolution = 6 bits, T _J = 140°C						-	-	46890	
		Resolution = 6 bits, T _J = 125°C						-	-	79000	